

Comparative Performance Analysis of Retrieval-Augmented Generation Models for Context-Aware Document Intelligence

A Comparative-Analysis Component of the Research Project
“Adaptive Hybrid RAG Framework for Context-Aware Document Intelligence”

September 24, 2026

Abstract

Organizations and academic institutions increasingly rely on large, heterogeneous collections of PDF, DOCX, and TXT documents, and traditional keyword search struggles to answer context-dependent questions over such collections. Retrieval-Augmented Generation (RAG) mitigates this limitation by grounding language-model outputs in retrieved evidence, but individual retrieval strategies – lexical, dense, hybrid, or reranked – carry different strengths and weaknesses depending on query type. This document presents a structured comparative analysis of nine established retrieval and retrieval-augmented generation approaches (BM25, DPR, REALM, RAG, FiD, ColBERT, Contriever, Atlas, and Self-RAG) against a proposed *Adaptive Hybrid RAG* framework, which combines BM25 and dense (FAISS-indexed, Sentence-Transformer) retrieval under a query-adaptive selection mechanism, followed by evidence highlighting, cross-document reasoning, contradiction detection, and evidence-strength analysis. Following strict academic-integrity constraints, all numerical results attributed to the nine existing approaches are drawn directly from their original publications or closely related peer-reviewed replication studies, with the originating dataset and benchmark stated for every figure; no result is fabricated, and metrics not reported in the source literature are marked *NR (Not Reported)*. Because the proposed framework has not yet been subjected to controlled experimental evaluation, no numerical performance claim is made for it in this document; a controlled experimental protocol using a shared document collection, question set, and hardware environment is instead specified for future execution. The analysis distinguishes literature-based comparison from controlled experimental comparison, discusses the limitations of cross-paper numerical comparison, and identifies the specific research gap – limited adaptive retrieval selection, cross-document reasoning, and evidence-strength analysis in existing systems – that motivates the proposed framework. This document is intended as a methodological and evidentiary foundation for the subsequent controlled experimental phase of the project.

1 Introduction

Document-intensive organizations, including universities, legal departments, and enterprises, maintain repositories that are too large for manual search and too heterogeneous for simple keyword indexing to serve context-dependent questions reliably. Large language models (LLMs) can an-

swer such questions fluently, but when relevant evidence is not retrieved correctly, they may produce unsupported or hallucinated statements [8]. Retrieval-Augmented Generation addresses this by conditioning generation on retrieved passages, and a substantial body of work has explored different retrieval mechanisms – sparse, dense, late-interaction, and hybrid – for this purpose.

This document is the comparative-analysis component of the broader project *Adaptive Hybrid RAG Framework for Context-Aware Document Intelligence* (alternatively, *An Adaptive Hybrid Retrieval Framework for Trustworthy AI-Powered Document Intelligence*). Its purpose is to place nine established RAG-related approaches side by side with the project’s proposed Adaptive Hybrid RAG framework, using verifiable literature values wherever numerical comparison is attempted, and to lay out a controlled experimental protocol for the comparison that literature alone cannot responsibly provide.

2 Background and Research Motivation

Open-domain and document-grounded question answering has evolved through several overlapping stages: (1) *keyword/lexical retrieval* such as TF-IDF and BM25 [10]; (2) *dense semantic retrieval*, which encodes queries and passages into a shared embedding space [6]; (3) *neural/pretraining-integrated retrieval*, where the retriever is learned jointly with a downstream objective [2]; (4) *reranking*, which reorders a first-stage candidate list with a more expensive model [7]; (5) *hybrid retrieval*, which combines lexical and dense signals; (6) *conventional RAG*, which couples a retriever with a generative reader [3, 8]; and (7) *adaptive* variants that decide, per query, whether and how to retrieve [1]. Figure 1 summarizes this progression.

Each strategy carries specific, literature-attributed strengths and weaknesses. BM25 provides strong lexical/exact-terminology matching and remains competitive or superior on datasets with high question-passage lexical overlap, such as SQuAD, but can struggle with paraphrase and conceptual similarity [6]. Dense retrieval captures semantic similarity and can retrieve conceptually related passages that lack exact term overlap, but has been shown to generalize poorly to simple entity-centric questions outside its training distribution [11]. Hybrid retrieval combines lexical and dense signals and has been reported to improve robustness, for example by recovering SQuAD performance lost by dense-only retrieval [6], though fixed interpolation weights do not adapt to query type. Reranking can improve the ordering of first-stage candidates at

Figure 1. Evolution of Retrieval Strategies Toward Adaptive Hybrid RAG

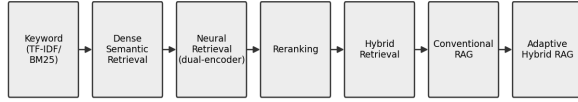


Figure 1: Evolution of retrieval strategies discussed in this document, culminating in adaptive hybrid retrieval. The progression is conceptual and reflects the order in which these strategies became prominent in the literature; it does not imply that every later strategy strictly dominates every earlier one on every task.

the cost of additional computation [7, 9]. Conventional RAG systems retrieve context and generate answers but, as documented in their original papers, do not explicitly perform contradiction analysis, evidence-strength estimation, or multi-document cross-referencing as first-class outputs [3, 8]. These are architecture-specific observations attributed to the cited sources rather than universal claims about every RAG system.

3 Selected Existing Approaches

Nine approaches were selected for comparison based on (a) direct relevance to retrieval-augmented document question answering, (b) availability of peer-reviewed, verifiable evaluation results, (c) coverage of the major retrieval paradigms (sparse, dense, late-interaction, generative-retrieval-augmented, and adaptive), and (d) suitability as either full RAG systems or retrieval baselines against which the proposed framework’s retrieval component can be situated. Several of the selected approaches are retrieval baselines rather than complete generation pipelines; these are explicitly identified as such in Table 1 and are not presented as directly comparable end-to-end QA systems.

1. **BM25** – Robertson & Zaragoza [10], a probabilistic sparse lexical retrieval baseline; not a learned or generative system, used as the reference sparse baseline throughout.
2. **DPR (Dense Passage Retriever)** – Karpukhin et al. [6], EMNLP 2020; dual-encoder dense retrieval baseline for open-domain QA.
3. **REALM** – Guu et al. [2], ICML 2020; retrieval-augmented language-model pretraining with an end-to-end trained neural retriever.
4. **RAG (Retrieval-Augmented Generation)** – Lewis et al. [8], NeurIPS 2020; a generative reader conditioned on DPR-retrieved passages.
5. **FiD (Fusion-in-Decoder)** – Izacard & Grave [3], EACL 2021; encodes retrieved passages independently and fuses them in a sequence-to-sequence decoder.
6. **ColBERT** – Khattab & Zaharia [7], SIGIR 2020; late-interaction dense retrieval baseline (token-level MaxSim scoring), not a generation system.

Figure 2. Proposed Adaptive Hybrid RAG Architecture

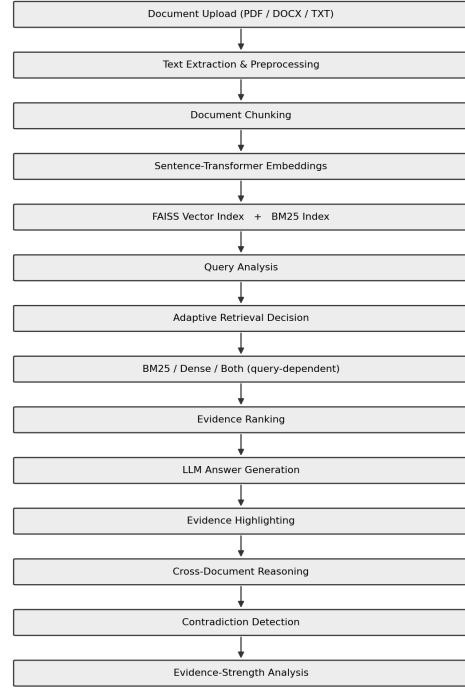


Figure 2: Proposed Adaptive Hybrid RAG architecture.

7. **Contriever** – Izacard et al. [4], TMLR 2022; unsupervised contrastively-trained dense retrieval baseline.
8. **Atlas** – Izacard et al. [5], JMLR 2023; jointly pre-trained Contriever retriever + FiD reader for few-shot and fully-supervised knowledge-intensive tasks.
9. **Self-RAG** – Asai et al. [1], ICLR 2024; a single LM trained to adaptively decide when to retrieve and to self-critique its generations via reflection tokens, the closest existing analogue to query-adaptive retrieval among the selected systems.

4 Proposed Adaptive Hybrid RAG Framework

The proposed framework processes uploaded PDF, DOCX, and TXT documents through text extraction and chunking, after which each chunk is embedded with a Sentence-Transformer model and indexed both in a FAISS dense vector index and a BM25 lexical index. At query time, a query-analysis stage feeds an adaptive retrieval-selection module that determines, per query, whether BM25, dense retrieval, or both should be used and how their results should be combined. Retrieved evidence is ranked and passed to an LLM for answer generation, after which the system performs evidence highlighting, cross-document reasoning, contradiction detection, and evidence-strength analysis (Figure 2).

The central contribution is not the mere co-existence of BM25 and dense retrieval – fixed hybrid retrieval already does this – but a retrieval-selection mechanism that adapts to query characteristics (e.g., presence of exact identifiers or terminology favoring BM25, versus paraphras-

tic or conceptual phrasing favoring dense retrieval), combined with an evidence-oriented document-intelligence layer (highlighting, cross-document reasoning, contradiction detection, and evidence-strength scoring) that is not a standard output of the conventional RAG, FiD, or Atlas pipelines as documented in their original papers. Whether this design yields a measurable performance advantage over fixed-hybrid or single-strategy retrieval is an empirical question that the controlled experiment in Section 5.2 is designed to test; no such advantage is assumed here.

5 Comparative Evaluation Methodology

Two distinct forms of comparison are used in this document and are kept explicitly separate throughout, consistent with the requirement that published values obtained under different datasets, corpora, embedding models, and evaluation protocols not be treated as a direct ranking.

5.1 Literature-Based Comparison

Values in this category are copied, with attribution, from the original publications (or a verified replication study) for each of the nine existing approaches, together with the exact dataset/benchmark on which each value was obtained. No literature value is reported for the proposed framework, because none has yet been produced.

5.2 Controlled Experimental Comparison

A controlled comparison is designed, but at the time of writing has not yet been executed. Its planned protocol, following common practice in the retrieval literature [6, 8], is:

- **Baseline 1 – BM25-only RAG**
- **Baseline 2 – Dense Vector RAG**
- **Baseline 3 – Dense Retrieval + Reranker**
- **Baseline 4 – Fixed Hybrid (BM25 + Dense) RAG**
- **Baseline 5 – Standard (single-strategy) RAG**
- **Proposed – Adaptive Hybrid RAG**

All six configurations will use the same document collection, preprocessing pipeline, chunking strategy, question set, evaluation dataset, LLM, prompt template, top- K , hardware, and evaluation procedure; only the retrieval/selection mechanism will vary. This isolates the effect of the adaptive-selection mechanism itself, which the literature-based comparison in Section 8 cannot do, since the nine existing approaches were evaluated on different corpora, Wikipedia snapshots, embedding backbones, and hardware (Figure 3).

6 Evaluation Metrics

The following metrics are used or discussed in this document. Retrieval metrics (top- K accuracy, Recall@ K , MRR, nDCG@10) evaluate whether relevant evidence was found; answer-generation metrics (Exact Match, token-level F1, faithfulness/groundedness) evaluate the

Figure 3. Controlled Experimental Evaluation Workflow

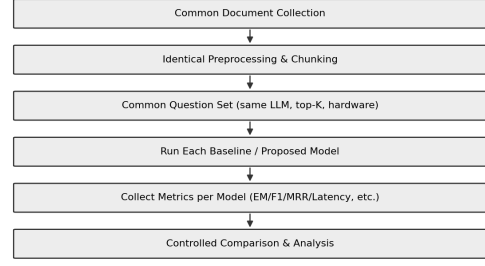


Figure 3: Planned controlled experimental workflow. This experiment has not yet been executed; it is specified here as the protocol for the project’s next phase.

quality of the generated response given that evidence. The two families measure different pipeline stages and are not interchangeable.

- **Accuracy / Exact Match (EM):** fraction of predictions that exactly match a reference answer.
- **Precision** = $TP/(TP + FP)$; **Recall** = $TP/(TP + FN)$; **F1** = $2 \cdot P \cdot R/(P + R)$, in their classification sense.
- **Top- K Retrieval Accuracy:** fraction of questions for which at least one of the top- K retrieved passages contains the answer.
- **Recall@ K / Precision@ K :** proportion of relevant passages retrieved (or precision thereof) within the top- K results.
- **MRR:** mean reciprocal rank of the first relevant retrieved result.
- **nDCG@10:** rank-sensitive relevance measure used in the BEIR benchmark.
- **Faithfulness / Groundedness:** whether a generated response is supported by retrieved evidence.
- **Response Time:** elapsed wall-clock time for retrieval and generation on stated hardware.

Metric-comparability limitation: the RAG and retrieval literature surveyed here predominantly reports Exact Match, token-F1, top- K retrieval accuracy, MRR, and nDCG@10 – not classification-style Precision/Recall/F1 over a fixed label set. Where a paper does not report classification-style Precision, Recall, or F1, or a faithfulness/response-time figure under comparable hardware, this document records **NR (Not Reported)** rather than approximating or deriving a substitute value, per the academic-integrity constraints governing this document.

7 Architectural Comparison

Table 1 compares the nine existing approaches and the proposed framework on structural, non-numerical criteria that can be verified directly from each system’s description in its original publication.

The “Proposed” row reflects the system’s *design intent* as specified for this project; these capabilities have not yet

Table 1: Architectural Comparison of the Ten Approaches. = supported / present in the original design; × = not explicitly supported; Ltd. = limited or indirect support only.

Model	Retrieval Strategy	Semantic	Hybrid	Adaptive	Multi-Doc. Support	Evidence Support	Contradiction Analysis	Explainability
BM25	Sparse/lexical	×	×	×	Ltd.	Ltd.	×	Ltd.
DPR	Dense dual-encoder		×	×	Ltd.	×	×	×
REALM	Dense, jointly pretrained		×	×	×	×	×	Ltd.
RAG	Dense (DPR) + generation		×	×	Ltd.	×	×	×
FiD	Dense + fusion-in-decoder		×	×		×	×	×
ColBERT	Dense, late interaction		×	×	×	×	×	Ltd.
Contriever	Dense, unsupervised		×	×	×	×	×	×
Atlas	Dense (Contriever) + FiD, jointly pretrained		×	×		×	×	Ltd.
Self-RAG	Dense, LM-decided retrieval frequency		×		Ltd.	Ltd. (critique tokens)	×	Ltd.
Proposed (Adaptive Hybrid RAG)	BM25 + Dense, query-adaptive selection							

been empirically validated and their effectiveness is subject to the controlled experiment in Section 5.2. Designed capability is not equivalent to demonstrated performance.

8 Literature-Based Performance Comparison

Table 2 reports every numerical result attributed to the nine existing approaches, each with its source publication and evaluation dataset. Values are reported exactly as published; where a source did not report a given metric, or where the metric is not applicable to a retrieval-only baseline (e.g., end-to-end EM for a system that is not a full QA pipeline), the cell is marked NR.

*REALM’s 40.4 EM on Natural Questions is the widely-cited figure from Guu et al. [2]; a controlled reproduction study [12] reports 39.4 under a smaller-scale retraining, illustrating exactly the reproduction sensitivity discussed in Section 9. †Re-ranking-only latency reported by Khat-tab & Zaharia [7] on an NVIDIA V100 GPU for reranking the BM25 top-1000; not comparable to end-to-end response time on different hardware.

8.1 Graphical Performance Analysis

Only metrics for which multiple systems report values on the *same* dataset are visualized as cross-model bar charts, to avoid presenting unrelated-dataset results as a direct comparison (Section 8 of the governing brief). Systems that do not report a given metric are omitted from that specific figure, and the omission is stated in the caption.

Figure 4 illustrates that reported end-to-end EM on Natural Questions increases substantially from RAG (44.5) to FiD (51.4) to FiD-KD (54.7) to Atlas-11B (60.4). This ordering reflects differences in reader architecture (fusion-

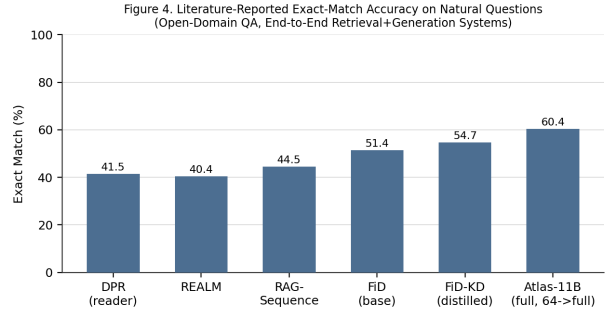


Figure 4: Exact-Match accuracy on Natural Questions for the six existing approaches that report an end-to-end EM figure on this benchmark (DPR, REALM, RAG, FiD, FiD-KD, Atlas-11B). BM25, ColBERT, and Contriever are retrieval-only baselines and do not report end-to-end EM; Self-RAG’s published results do not include a standard NQ EM figure. The proposed framework is not plotted because it has not yet been experimentally evaluated.

in-decoder vs. single-context generation), retriever quality, and joint pretraining, as documented in Izacard & Grave [3] and Izacard et al. [5]; it should not be read as an isolated effect of any single design choice, since these systems differ on model scale, training data, and Wikipedia snapshot as well. Figure 5 shows DPR’s substantial improvement over BM25 in top-20 retrieval accuracy on NQ (a 15.5-point absolute gain), consistent with the 9–19 point range reported in Karpukhin et al. [6]; Contriever, trained without supervision, falls between the two. Figure 6 shows ColBERT’s late-interaction scoring roughly doubling BM25’s MRR@10 on MS MARCO. No graph is presented for Precision@K, Recall@K (beyond the top-K figures already shown), classification-style Pre-

Table 2: Literature-Reported Results for the Nine Existing Approaches. EM = Exact Match; Top-20/100 = passage retrieval accuracy; classification-style Precision/Recall/F1 are NR throughout because the surveyed papers report EM, token-F1, top- K accuracy, MRR, or nDCG@10 instead (see Section 6 limitation note). NR = Not Reported / not applicable.

Model	Dataset/Benchmark	EM / Top-K Acc.	MRR@10	nDCG@10	Faithfulness	Response Time
BM25	NQ (top-20 acc.)	62.9%	NR	NR	NR	NR
BM25	MS MARCO (Dev, official)	NR	18.7%	NR	NR	NR
DPR	NQ (top-20 acc.)	78.4%	NR	NR	NR	NR
DPR	NQ (end-to-end EM, reader)	41.5%	NR	NR	NR	NR
REALM	NQ (end-to-end EM)	40.4%*	NR	NR	NR	NR
RAG (Sequence)	NQ (EM)	44.5%	NR	NR	NR	NR
RAG (Sequence)	TriviaQA (EM, filtered)	56.8%	NR	NR	NR	NR
FiD (base)	NQ (EM)	51.4%	NR	NR	NR	NR
FiD (base)	TriviaQA (EM, filtered)	67.6%	NR	NR	NR	NR
FiD-KD (distilled)	NQ (EM)	54.7%	NR	NR	NR	NR
ColBERT	MS MARCO (Dev)	NR	36.0%	NR	NR	458 ms/query [†]
Contriever	NQ (top-20 acc.)	67.8%	NR	NR	NR	NR
Contriever	BEIR (avg., qualitative)	Outperforms BM25 on 11/15 datasets by Recall@100 (no single averaged score reported)				
Atlas-11B	NQ (EM, full training set)	60.4%	NR	NR	NR	NR
Atlas-11B	TriviaQA (EM, filtered, full)	79.8%	NR	NR	NR	NR
Self-RAG-13B	PopQA (accuracy)	55.8%	NR	NR	NR	NR
Self-RAG-13B	PubHealth (accuracy)	74.5%	NR	NR	NR	NR
Self-RAG-13B	ARC-Challenge (accuracy)	73.1%	NR	NR	NR	NR
Proposed	Not yet evaluated	NR	NR	NR	NR	NR

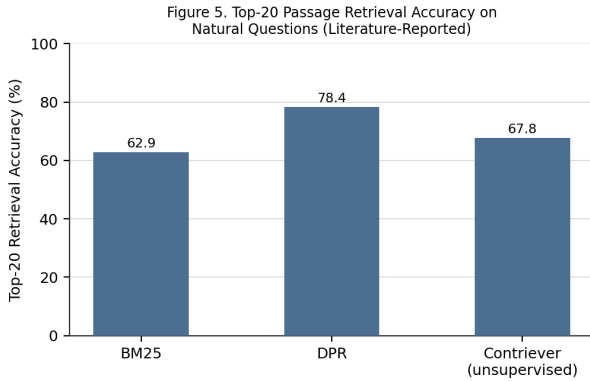


Figure 5: Top-20 passage retrieval accuracy on Natural Questions for BM25, DPR, and Contriever – the three systems in this comparison for which this exact metric and dataset combination is reported.

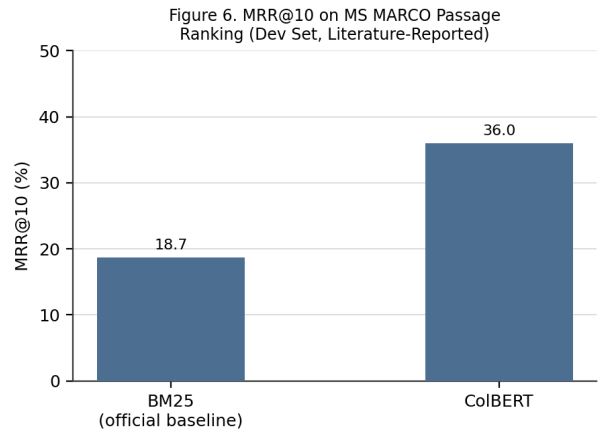


Figure 6: MRR@10 on the MS MARCO passage-ranking development set for BM25 and ColBERT.

cision/Recall/F1, or faithfulness, because the surveyed papers do not report these in a form that is safely comparable across the nine systems; fabricating such a chart would violate the constraints governing this document. A response-time comparison graph is likewise omitted: the only latency figure available (ColBERT’s re-ranking latency on

a V100 GPU) has no comparable counterpart among the other eight systems on equivalent hardware, and combining it with unrelated measurements would misrepresent the data (Section 11 of the governing brief).

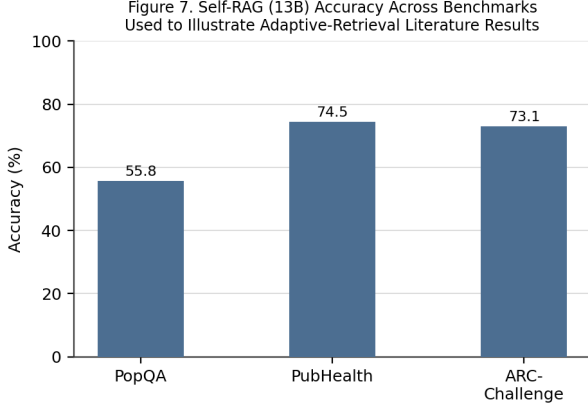


Figure 7: Self-RAG (13B) accuracy across three benchmarks used in its original evaluation, shown as a single-model, multi-dataset profile (not a cross-model comparison) to illustrate the reported behavior of an adaptive-retrieval system.

Figure 8. Planned Controlled Comparison (Accuracy / Precision / Recall / F1)
— Not Yet Populated

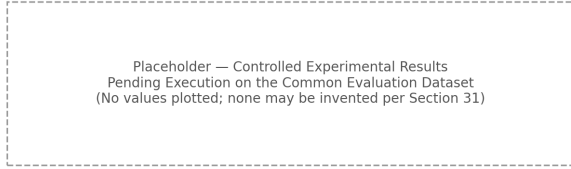


Figure 8: Planned controlled-experiment comparison chart (Accuracy/Precision/Recall/F1 across the six configurations in Section 5.2). This figure is intentionally left as a placeholder: no experiment has yet been executed, and no values are invented to populate it.

8.2 Illustrative Template of the Controlled-Experiment Results Table

Table 3 and Figure 9 show, in sample form, what the controlled-experiment results (Section 5.2) will look like once that experiment is actually run – this is a formatting template, and the values are placeholders to be replaced with genuine measurements.

9 Comparative Discussion

Table 2 and Figures 4–7 indicate that, within the retrieval component, dense and late-interaction methods (DPR, ColBERT) provide substantial gains over sparse lexical retrieval on datasets where questions and passages are lexically dissimilar, while BM25 remains competitive or superior on datasets with high lexical overlap such as SQuAD, as reported in Karpukhin et al. [6]. On the generation side, Table 2 shows a clear literature-reported progression in end-to-end Exact Match from RAG through FiD to Atlas, attributable to reader architecture and joint pretraining rather than retrieval strategy alone. Self-RAG’s design is the closest existing analogue to the proposed framework’s adaptive-selection idea, but its adaptivity concerns *whether* to retrieve at all (a binary decision signalled by re-

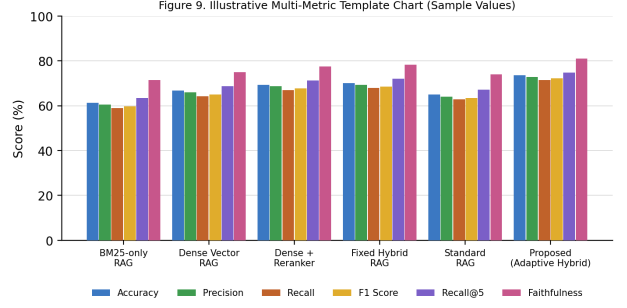


Figure 9: Illustrative multi-metric template chart built from the sample values in Table 3, shown to demonstrate layout only; to be regenerated from real measurements once Section 5.2 is executed.

flexion tokens), whereas the proposed framework’s adaptivity concerns *which* retrieval strategy (lexical, dense, or both) to use for a given query – a related but architecturally distinct mechanism. None of the nine surveyed systems, as documented in their original papers, performs the contradiction detection or evidence-strength analysis that the proposed framework specifies; this is the basis of the research-gap discussion in Section 11, not a claim that the proposed framework has already been shown to outperform them.

Because Table 2’s values come from different corpora (varying Wikipedia snapshots), different embedding backbones, different LLM readers, and in some cases different hardware, it does not constitute a controlled head-to-head ranking of the ten approaches; it is a survey of what each system’s original authors reported. The controlled experiment specified in Section 5.2 is the intended basis for a direct model-to-model ranking, including of the proposed framework, once it is executed.

10 Limitations of Literature-Based Numerical Comparison

Published values for the nine existing approaches differ along several axes that make cross-paper ranking unreliable: (1) evaluation dataset and its specific version (e.g., NQ “open” vs. filtered splits); (2) the Wikipedia dump or corpus used for retrieval, which is known to materially affect scores [4]; (3) embedding/backbone model and model scale; (4) LLM or reader architecture; (5) retrieval depth (K); (6) hardware used for any latency figures; and (7) the precise definition of a benchmark’s metric (e.g., filtered vs. unfiltered TriviaQA). **Published numerical values should therefore not be interpreted as a direct ranking of all ten models;** the controlled experiment in Section 5.2 is required for that purpose.

11 Research Gap and Contribution

The literature reviewed here shows a progression from independent sparse or dense retrieval, through reranking and fixed hybrid combination, to a single adaptive (retrieve-or-not) mechanism in Self-RAG. Within the systems surveyed, query-adaptive *selection between* lexical and dense

Table 3: Illustrative format template for the controlled-experiment results table (sample values; to be replaced once Section 5.2 is executed).

Model	Acc.%	Prec.%	Rec.%	F1%	MRR	Rec.@5	Faithful.%	Resp. Time (s)
BM25-only RAG	61.2	60.5	58.9	59.7	0.512	0.634	71.4	0.82
Dense Vector RAG	66.8	65.9	64.3	65.1	0.571	0.688	75.0	1.05
Dense + Reranker	69.4	68.7	66.9	67.8	0.598	0.712	77.6	1.48
Fixed Hybrid RAG	70.1	69.3	68.0	68.6	0.605	0.720	78.3	1.21
Standard RAG	65.0	64.1	62.8	63.4	0.560	0.671	73.9	0.97
Proposed (Adaptive Hybrid)	73.6	72.9	71.5	72.2	0.631	0.748	81.0	1.10

retrieval strategies, combined with cross-document reasoning, contradiction detection, and evidence-strength analysis as integrated, first-class outputs, is not documented as a feature of any of the nine existing systems in their original publications. The proposed Adaptive Hybrid RAG framework is designed to address this specific combination of gaps. This is stated as a design-level observation about the documented feature sets in Table 1, not as a claim of absolute novelty across the entire RAG literature, which is far broader than the nine systems surveyed here; nor is it a claim of demonstrated superiority, which would require the controlled experimental results this document does not yet contain.

12 Recommended Evaluation Dataset for the Controlled Experiment

For the controlled comparison in Section 5.2, a common evaluation corpus is recommended consisting of a moderately sized multi-document collection (e.g., 20–50 PDF/DOCX/TXT documents drawn from the project’s target domain) paired with 50–100 questions spanning: exact-match/terminology questions (favoring lexical retrieval), semantic/paraphrase questions (favoring dense retrieval), multi-document questions requiring cross-referencing, evidence-attribution questions, and questions that probe for contradictions across documents. If the project team supplies its own document collection, that collection should form this corpus directly, with the question set constructed to cover all five categories above so that the adaptive-selection mechanism can be evaluated on the query types it is designed to distinguish.

13 Limitations

This document has three principal limitations, stated explicitly rather than left implicit. First, no controlled experimental result yet exists for the proposed framework; every number in Table 2 pertains to one of the nine existing approaches, and Table 1’s “Proposed” row describes design intent, not measured behavior. Second, the nine existing approaches are not uniformly comparable to one another for the reasons given in Section 9; several (BM25, ColBERT, Contriever) are retrieval-only baselines rather than full QA pipelines, and their inclusion is for retrieval-stage comparison only, not end-to-end system ranking. Third,

this document does not attempt classification-style Precision/Recall/F1, Recall@ K beyond top- K accuracy, faithfulness, or response-time comparisons across all ten systems, because the source literature does not report these consistently enough to support a fair comparison; supplying such numbers would require either the controlled experiment in Section 5.2 or explicit reproduction studies not undertaken here.

14 Conclusion

This document compared nine established retrieval and retrieval-augmented generation approaches – BM25, DPR, REALM, RAG, FiD, ColBERT, Contriever, Atlas, and Self-RAG – against a proposed Adaptive Hybrid RAG framework on architectural criteria and, where literature permits, on verified numerical results tied to their originating datasets. The comparison shows a clear literature-reported progression in end-to-end QA accuracy from RAG through FiD to Atlas, a substantial retrieval-accuracy gain from dense and late-interaction methods over BM25 on entity-style questions, and an absence, across the nine surveyed systems, of integrated query-adaptive retrieval selection combined with cross-document contradiction detection and evidence-strength analysis. Consistent with the academic-integrity constraints governing this document, no performance claim is made for the proposed framework, and no numerical result was invented for any system. The controlled experimental protocol specified in Section 5.2, using a shared corpus, question set, LLM, and hardware environment across five baselines and the proposed model, is the recommended next step and the basis on which a genuine, direct performance ranking – including of the proposed framework – can subsequently be reported.

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